Data Analyst Capstone Project

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November 11, 2026



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EXECUTIVE SUMMARY



We studied the Stack Overflow developer survey (65,437 respondents) plus job and salary data.

- Top languages today: JavaScript, HTML/CSS, Python, SQL, TypeScript
- Python becomes #1 next year; Rust and Go enter the top 10
- PostgreSQL is the top database now and next year; SQLite and Supabase rise
- Most job postings: Washington DC, Detroit, Seattle, New York
- Swift pays the most ($130,801); Python and C++ follow

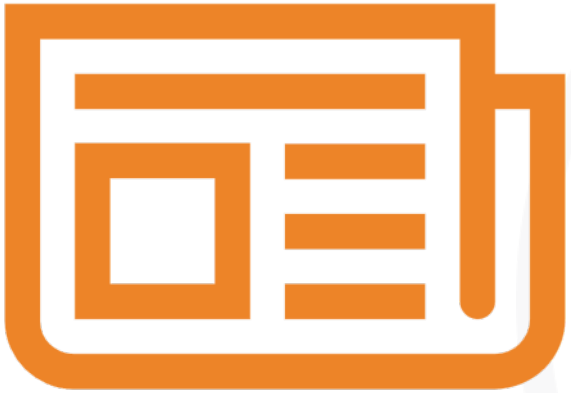
INTRODUCTION



Why this report, who it is for, and why it matters

- Purpose: show which technologies developers use today and want to use next year
- Audience: students, job seekers, hiring managers and IT leaders
- Value: helps people decide which languages and databases to learn
- Questions: Which languages and databases lead? Where are the jobs? Which languages pay most? Who are the developers?

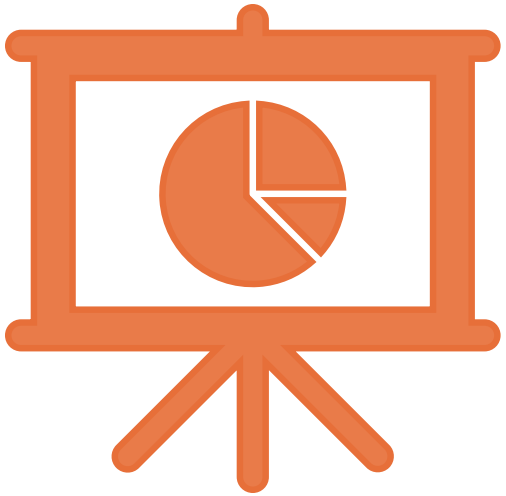
METHODOLOGY



Data sources, data collection and data wrangling steps

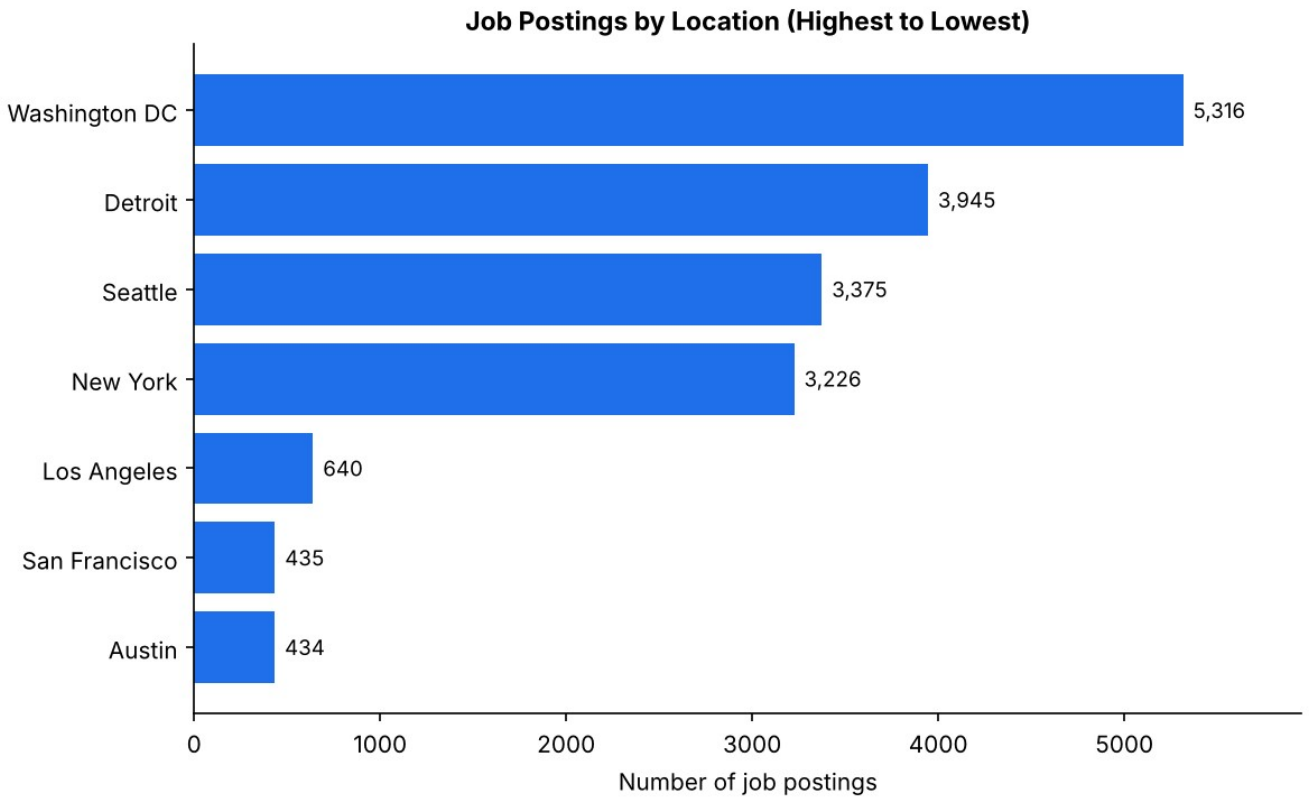
- Sources: Stack Overflow Developer Survey (65,437 respondents), a Jobs API (7 locations), and a web-scraped table of languages and salaries
- Collection: Python (requests, BeautifulSoup, pandas); saved as job-postings.xlsx and popular-languages.csv
- Wrangling: split multi-answer columns, counted each technology, ranked the top 10
- Compared "worked with" (current year) to "want to work with" (next year)
- Visuals: bar charts and a 3-tab Excel dashboard

Results



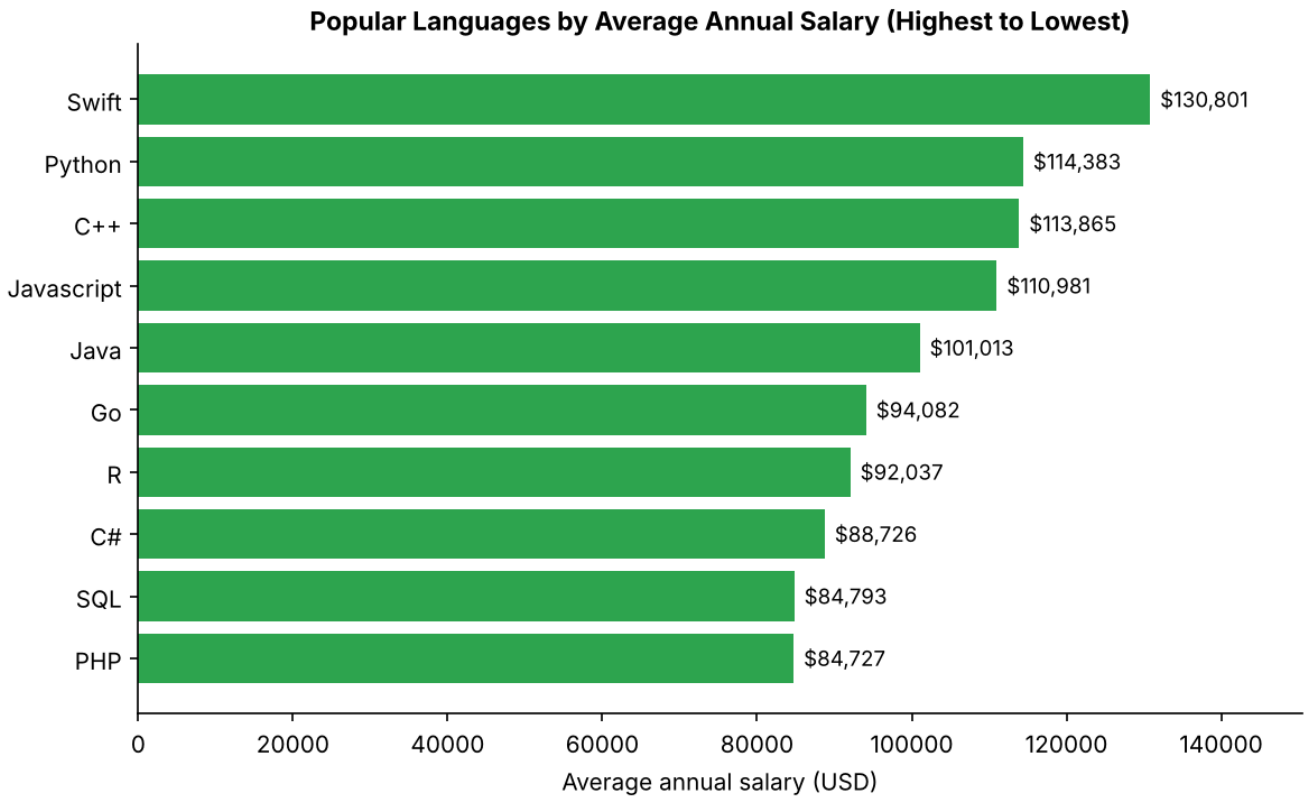
Job postings, language salaries and technology trends.

JOB POSTINGS



- Washington DC has the most job postings (5,316).
- Detroit, Seattle and New York each have more than 3,000.
- Los Angeles, San Francisco and Austin have far fewer (under 700).
- Source: Jobs API from the course (7 locations).

POPULAR LANGUAGES

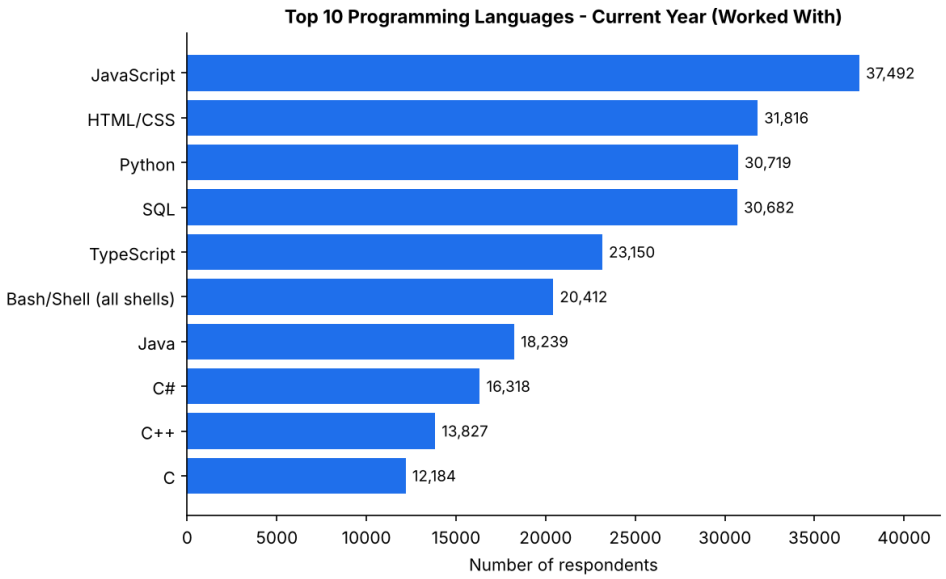


- Swift pays the most: $130,801 a year on average.
- Python ($114,383), C++ ($113,865) and JavaScript ($110,981) follow.
- SQL and PHP are lowest in this list, at about $84,700.
- Source: web-scraped table of popular languages.

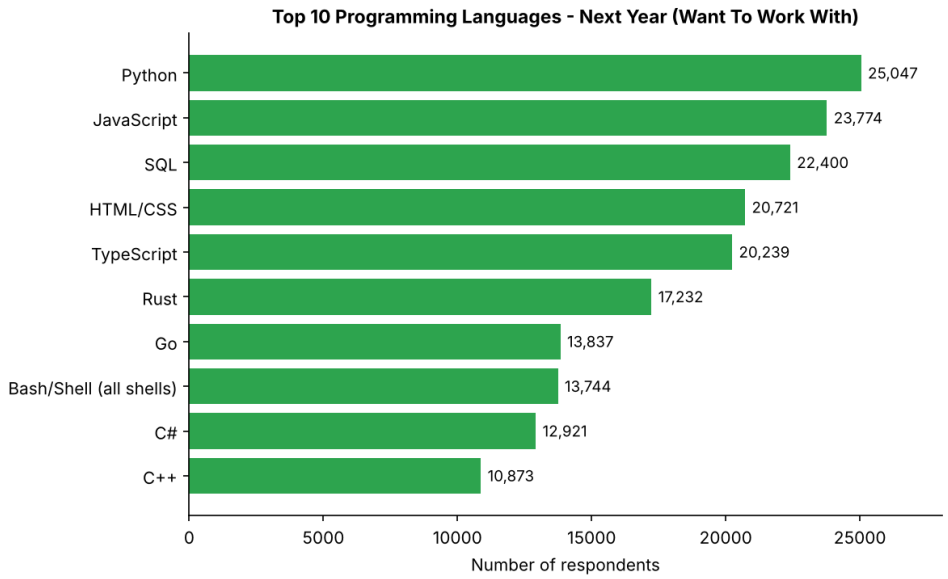
PROGRAMMING LANGUAGE TRENDS

JavaScript leads today; Python moves to #1 next year, and Rust and Go enter the top 10.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript is #1 today (37,492); Python is #1 next year (25,047)
- Rust and Go enter the next-year top 10; Java and C drop out
- Web languages and SQL stay strong in both years

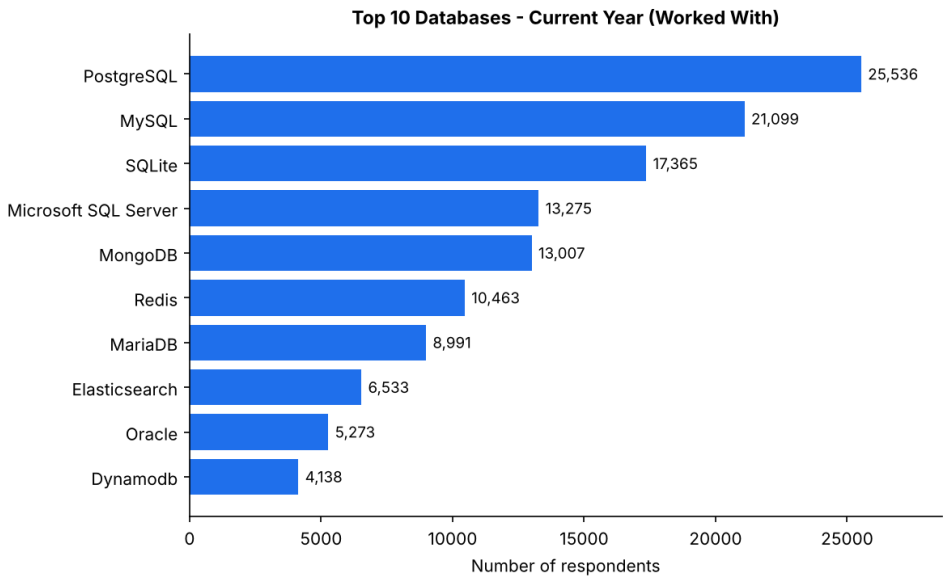
Implications

- Python and SQL are the safest skills to learn
- Rust and Go are good extra skills to stand out
- Expect less demand for Java and C

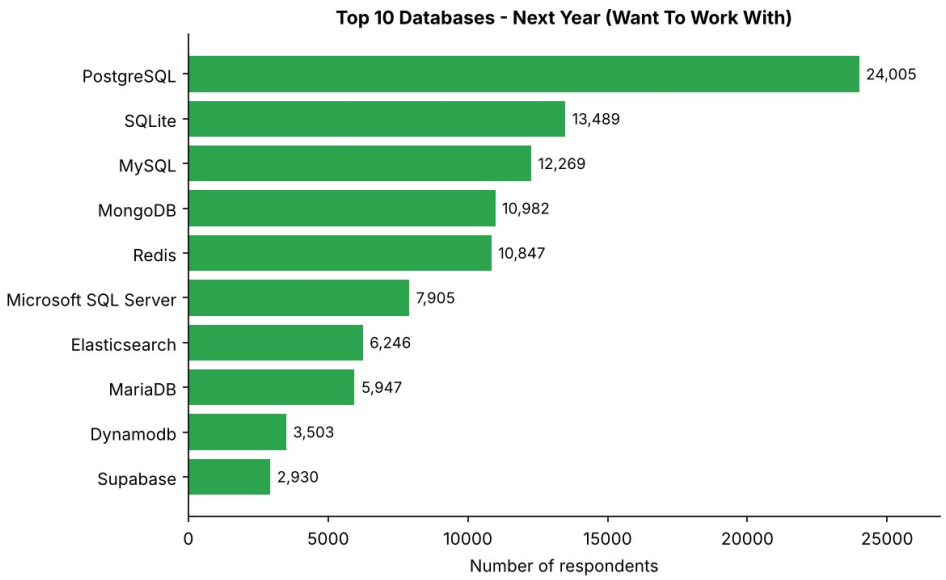
DATABASE TRENDS

PostgreSQL leads now and next year; SQLite moves to #2 and Supabase enters the top 10.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is #1 in both years by a wide margin
- MySQL falls from #2 to #3; SQL Server falls from #4 to #6
- Oracle leaves the top 10; Supabase enters

Implications

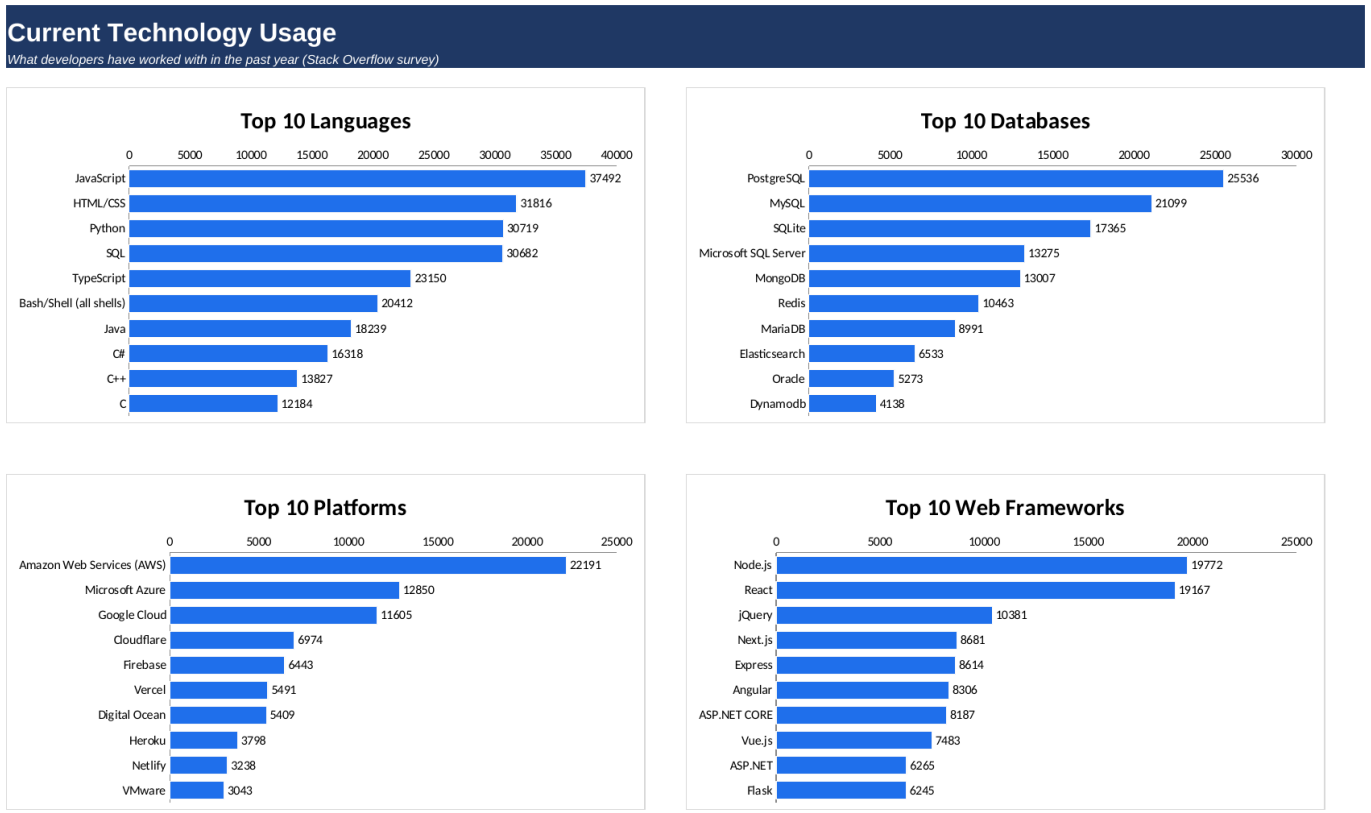
- Learn PostgreSQL first
- Older enterprise databases are losing interest
- Easy, newer databases like SQLite and Supabase are growing

DASHBOARD



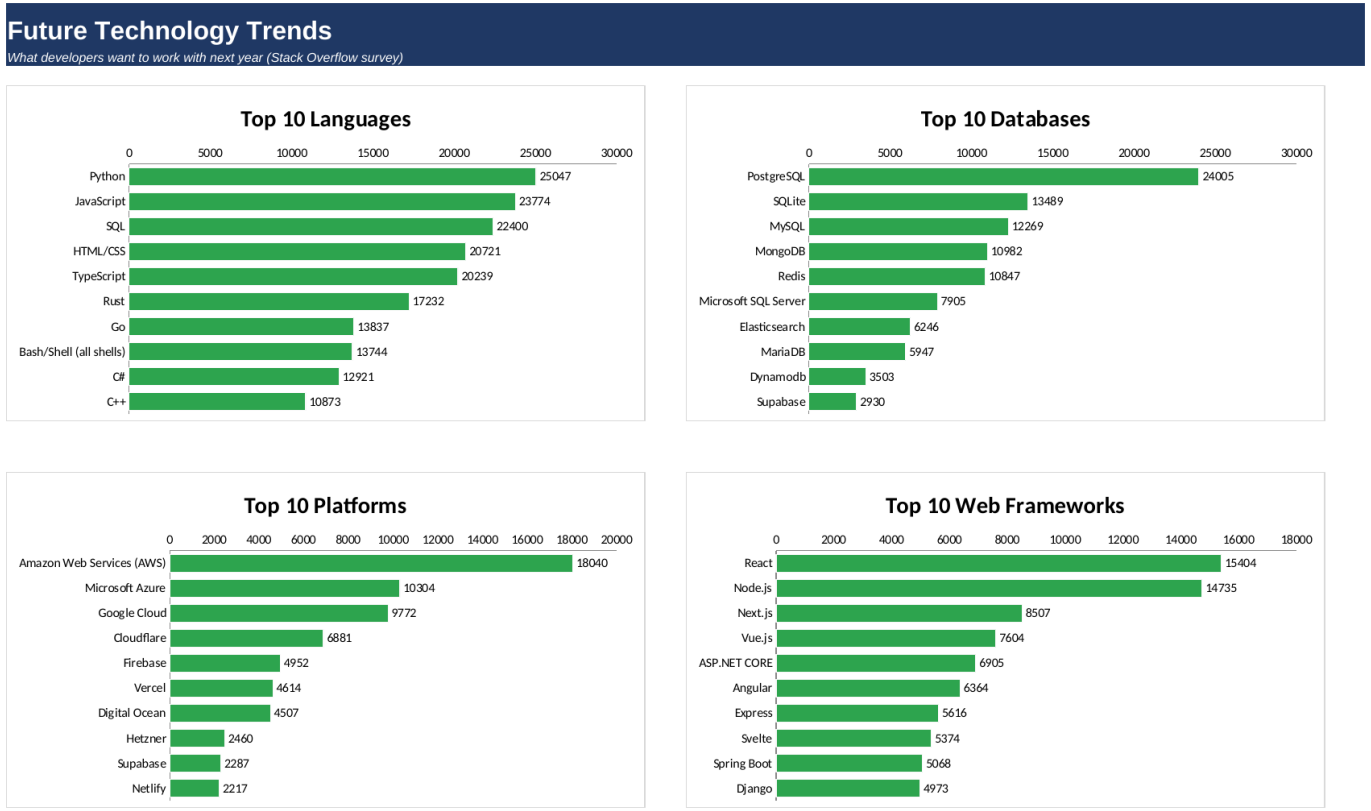
A 3-tab Excel dashboard built from the survey data.

DASHBOARD TAB 1: Current Technology Usage



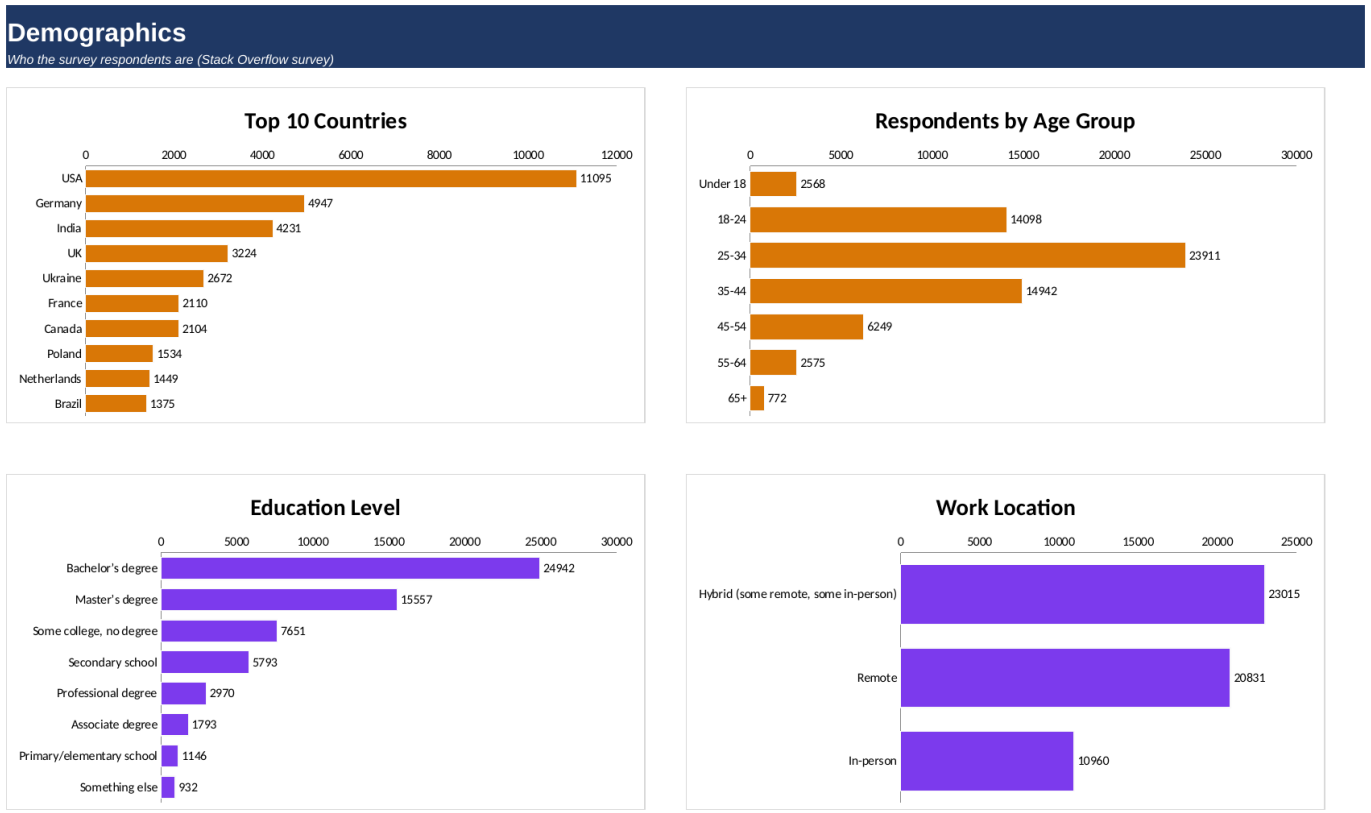
- Dashboard Insight
- JavaScript is the most used language (37,492), then HTML/CSS, Python and SQL.
- PostgreSQL is the top database, ahead of MySQL and SQLite.
- AWS leads platforms (22,191); Node.js and React lead web frameworks.

DASHBOARD TAB 2: Future Technology Trends



- Dashboard Insight
- Python moves to #1 next year; Rust and Go enter the top 10.
- PostgreSQL stays #1; SQLite moves up to #2.
- React moves ahead of Node.js; Supabase, Svelte and Hetzner appear.

DASHBOARD TAB 3: Demographics



- Dashboard Insight
- The USA has the most respondents (11,095), then Germany and India.
- Ages 25-34 are the largest group; a Bachelor's degree is the most common.
- Most people work hybrid (23,015) or remote (20,831).

DISCUSSION



- Developers use web languages and SQL most today, but want Python even more next year.
- Cloud (AWS, Azure, Google Cloud) and open-source databases lead; PostgreSQL is the clear favorite.
- React and Node.js lead web frameworks; newer tools like Svelte and Supabase are rising.
- Most respondents are 25-34, hold a Bachelor's degree, and work hybrid or remote.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Python, JavaScript and SQL are core skills now and next year
- PostgreSQL leads databases; SQLite and Supabase are growing
- Most developers are 25-34, hold a Bachelor’s degree, and work hybrid or remote

Implications

- Learn Python and SQL first, then add JavaScript or TypeScript
- Add PostgreSQL and a cloud platform such as AWS
- Job seekers can target Washington DC, Detroit, Seattle, New York and remote roles

CONCLUSION



- Python, JavaScript, HTML/CSS and SQL are today's core languages; Python leads next year.
- PostgreSQL is the top database now and next year.
- AWS leads cloud platforms; React and Node.js lead web frameworks.
- Python and SQL give the best mix of demand and pay.

APPENDIX



- Capstone_Dashboard.xlsx holds the data tables behind all dashboard charts.
- Platforms and web frameworks (top 10, current and next year) are shown on the dashboard slides.
- Files used: survey data (65,437 rows), job-postings.xlsx, popular-languages.csv.